An Exact Prime Classification for OEIS A308090

Qichao Wang
Hebei University of Technology
Daoyu Dong
University of Electronic Science and Technology of China

October 26, 2026

Abstract

The factorial gcd sequence in OEIS A308090 equals $n+1$ exactly when $n+1$ is a prime at least 5, and equals 1 otherwise. This proves the recorded converse conjecture and the proposed equality with a prime-indicator sequence, with its stated exceptional index. The proof uses Wilson's theorem and a complete treatment of the composite factorial exception.

1 Complete classification

Let

$$a(n) = \gcd(2^n + n!, 3^n + n!, n+1), \qquad n \geq 1.$$

The official entry [1] conjectures that $a(n) = n+1$ implies primality of $n+1$ and observes agreement, except at $n = 2$, with A090585.

Lemma 1. *If $m > 4$ is composite, then $m \mid (m-1)!$.*

Proof. If $m = ab$ with $1 < a < b < m$, both distinct factors occur in the factorial. The remaining case is $m = p^2$ for a prime p: if p is odd, the distinct factors p and $2p$ lie below p^2, so their product is divisible by p^2. The only excluded square is 4. □

Theorem 1. *For every $n \geq 1$,*

$$a(n) = \begin{cases} n+1, & n+1 \text{ is prime and } n \geq 4, \\ 1, & \text{otherwise.} \end{cases}$$

Proof. Write $m = n+1$. If $m \geq 5$ is prime, Wilson's theorem gives $n! \equiv -1 \pmod{m}$, and Fermat's theorem gives $2^n \equiv 3^n \equiv 1 \pmod{m}$. Both sums are divisible by m, so the gcd is m.

If $m > 4$ is composite, the lemma gives $n! \equiv 0 \pmod{m}$. Hence

$$a(n) = \gcd(2^n, 3^n, m) = 1.$$

Direct evaluation at $m = 2, 3, 4$ gives 1 in each case, completing the proof. □

Thus the conjectured converse holds without exceptions. A090585 has value $n+1$ when $n+1$ is an odd prime and 1 otherwise, so the entries agree except at $n = 2$.

The official comment concerning either single sum requires care: divisibility of one sum alone is not a primality criterion. For example, at $n = 7$, $8 \mid 2^7 + 7!$ although 8 is composite. The theorem concerns the joint gcd and establishes the precise valid statement.

2 Validation and scope

The standard-library verifier compares all 99 displayed official terms and independently computes the defining modular gcd for every $1 \leq n \leq 5000$. It also checks the supplied 100-row table. Modular arithmetic avoids materializing $n!$ or the exponential powers, but trial-division primality testing is not asserted to have polynomial cost in the bit length of n. The proof determines every index, independently of the finite controls. This is a proof of statements recorded in the cited entry, without a claim of literature priority or OEIS acceptance.

References

[1] OEIS Foundation Inc., A308090, https://oeis.org/A308090, accessed October 26, 2026.